SINGAPORE UNIVERSITY OF SOCIAL SCIENCES

MTH220

Special Resit Examination – June 2024

Statistical Methods and Inference

Thursday, 13 June 2024 **01:00 pm - 03:00 pm**

Time allowed: 2 hours

INSTRUCTIONS TO STUDENTS:

1. This examination contains **FOUR (4)** questions and comprises **SEVEN (7)** printed pages (including cover page).
2. You must answer **ALL** questions.
3. All answers must be written in the answer book. Marks will only be awarded when full working is shown.
4. This is an **Open-book** examination.

At the end of the examination.
Please ensure that you have written your examination number on each answer book used.

Failure to do so will mean that your work cannot be identified.
If you have used more than one answer book, please tie them together with the string provided.

THE UNIVERSITY RESERVES THE RIGHT NOT TO MARK YOUR SCRIPT IF YOU FAIL TO FOLLOW THESE INSTRUCTIONS.

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(Full marks: 100)

Question 1

The HR department in a large company wants to investigate if there is a difference in job satisfaction between males and females in the business development department. A random sample of 40 males and 45 females was selected and scored using a job satisfaction questionnaire. The job satisfaction scores are out of 100.

Question 1a

The men scored an average of 67 points, with a sample variance of 9, while the women scored an average of 69 points, with sample variance of 10. Assume that the scores from both groups are normally distributed with the same population variance.

Question 1a(i)

Formulate the null and alternative hypotheses.

(2 marks)

Question 1a(ii)

Calculate the pooled variance.

(3 marks)

Question 1a(iii)

At 1% level of significance, apply a two-sample t-test of equal variance to determine if there a statistically significant difference in the average scores between the two groups of people and comment on the results.

(10 marks)

Question 1b

The HR expanded the survey to increase the survey sample size to 300 females. Suppose that among the surveyed females, 231 of them provided feedback that having flexible working hours will improve their job satisfaction.

Compute a 95% CI for the true population of females who feels that flexible working hours will improve their job satisfaction.

(10 marks)

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Question 2

A nutritionist wants to compare the effects of three different diet plans (Diet A, Diet B, and Diet C) on weight loss. The nutritionist randomly assigns 36 participants into three groups, with 12 participants in each group. After six months on their respective diets, the weight loss (in kg) for each participant is recorded. Assume that the weight loss data in each group follows a normal distribution. The weight loss data for the three groups are as follows:

Diet	Weight loss (kg)											
A	10	12	8	9	11	14	13	11	12	9	12	11
B	10	7	8	6	6	7	13	8	6	10	7	8
C	1	3	6	7	4	1	3	5	4	6	17	3

Question 2a

Describe the purpose of the one-way ANOVA and it's limitations.

(5 marks)

Question 2b

Is there a significant difference in the mean weight loss among the three diet plans? Apply a one-way ANOVA at a 5% significance level to compare the mean weight loss across the three diet groups. Show all computations clearly, and direct output from computer is not accepted.

Question 2b(i)

Formulate the null and alternative hypotheses.

(2 marks)

Question 2b(ii)

Compute the SSE, SSTR and SST based on the dataset above.

(12 marks)

Question 2b(iii)

Draw your conclusions with either the use of p-value or critical value and comment on the results.

(6 marks)

Question 3a

Suppose you are a quality control manager at a beverage company. The bottling machine is expected to fill each bottle with an average of 330 ml of liquid, with a standard deviation of 5 ml. To ensure that the machine is functioning properly, you decide to take a sample of 100 bottles and measure their liquid content.

Question 3a(i)

According to the Central Limit Theorem, what can you expect to be the distribution of the sample means?

(3 marks)

Question 3a(ii)

Calculate the mean and standard deviation of the sample means.

(5 marks)

Question 3a(iii)

Solve and estimate the sample size n that should be taken if approximately 5% of the sample means are more than 331 ml.

(5 marks)

Question 3a(iv)

Explain the importance of the Central Limit Theorem in this quality control scenario.

(5 marks)

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Question 3b

Researchers conducted a study to examine the relationship between smoking status (non-smoker, occasional smoker, regular smoker) and the occurrence of respiratory symptoms (none, mild, severe) among a randomly selected sample of individuals. The data collected from 424 participants is summarized in the table:

	None	Mild	Severe	**Total**
Non-smoker	92	67	38	**197**
Occasional Smoker	51	41	22	**114**
Regular Smoker	33	53	27	**113**
Total	**176**	**161**	**87**	**424**

Is there evidence of a statistically significant association between smoking status and the occurrence of respiratory symptoms? Apply a chi-square test of independence using a 5% significance level to analyze the relationship between smoking status and the severity of respiratory symptoms.

Question 3b(i)

Formulate the null and alternative hypotheses and compute the expected frequencies assuming independence between smoking status and occurrence of respiratory symptoms.

(9 marks)

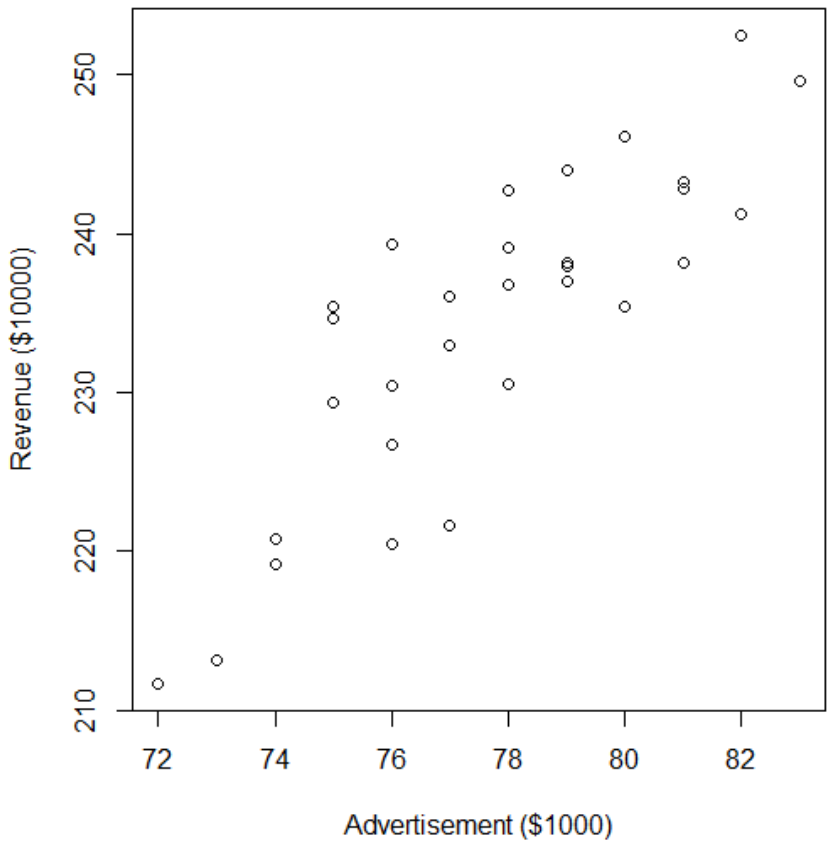
Question 3b(ii)

Compute the test statistic and comment on the results.

(8 marks)

Question 4a

A company wants to analyze the relationship between advertising expenses and sales revenue. The company collected data from 30 different advertising campaigns, recording the advertising expenses (x, in units of \$1k) and the corresponding sales revenue (y, in units of \$10k), as shown in the plot below. The company wants to build a regression model to predict sales revenue based on advertising expenses.



Question 4a(i)

A statistician was working on this dataset. This dataset was saved as "dat", and "dat" comprises of the x and y. Here is a screenshot of the R script:

```
Console   Terminal ×   Background Jobs ×
R 4.2.0 · ~/
> sum(dat$x)
[1] 2331
> sum(dat$y)
[1] 7027
> sum(dat$x^2)
[1] 181351
> sum(dat$y^2)
[1] 1648991
> sum(dat$x*dat$y)
[1] 546706
> lm(y~x,data=dat)

Call:
lm(formula = y ~ x, data = dat)

Coefficients:
(Intercept)            x
     [redacted]      [redacted]

>
```

Explain, the meaning of the codes above and what do they do. Use the output from R above to compute $\bar{x}, \bar{y}$ and write down the values of $\sum xy, \sum x^2$.

(5 marks)

Question 4a(ii)

Some of the results from the lm formula has been omitted. From the information provided in the numbers above, determine the equation of the least squares linear regression line.

(6 marks)

Question 4a(iii)

One of the staff from the marketing team wants to know the expected revenue when they spend $40,000 on advertising expenses this year. Compute the expected revenue and comment on the appropriateness of your answer.

(4 marks)

----- END OF PAPER -----